Joshua Prettyman, Ph.D.

I build scalable software solutions to meet business needs

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Personal Statement

I built an ML-powered SaaS platform that increased productivity by 20×, taking internal tooling from spreadsheets to a marketable product: full-stack, handling everything from NLP to cloud services to UX.

As a freelancer I have built good working relationships with my clients and consider business problems from every angle, leveraging my background in data, development and SEO to build scalable and profitable solutions.

Technical Skills

- Web development with Shopify, Sanity, Vercel, React, Liquid, JavaScript, TypeScript.
- Full stack development with Bash, Python, Google Cloud Platform (GCP), SQL.
- MLOps, RAG, NLP, OpenAI/Anthropic API integrations and fine-tuning.
- Data engineering (AWS, GCP, Databricks, dbt), ETL & ELT, analysis, dashboards (Tableau, Retool).
- Git, GitHub, Docker, package management, CI/CD, async, REST, GraphQL.
- Experience with C++, MATLAB, scientific computing, geospatial data, PyTorch.

Professional Experience

Data and integrations Architect — joshuaprettyman.com/freelance *Jan. 2025 to Present*

Delivered projects including: Website development, Technical SEO, Dashboard Creation, Pipeline Automation (dbt, Databricks, Apache), Database Setup, Data Engineering and Data Analysis.

- I work closely with all my clients to pick apart business problems and work out plans towards solutions.
- On web development projects I apply my years of SEO knowledge to every decision.
- I estimate time/cost then implement and document the agreed spec with a continuous feedback cycle.

Data Scientist — Blink SEO / Macaroni Software *Nov. 2021 to Dec. 2024*

Built *Macalytics*, a full-stack ML SaaS platform, from scratch. Increased team productivity by 20× and enabling year-long SEO campaigns to be delivered in weeks.

- Scaled from solo developer (me, first 18 months) to a four-person Agile team, introducing sprint-based delivery, user stories and code review as the team grew.
- Built interactive dashboards linked to Google Search Console, Google Analytics and Google Ads to identify and prioritise revenue opportunities.
- Integrated LLMs to generate immediate-impact, SEO-tuned suggestions for content gaps.
- Created Shopify integrations to instantly respond to
- Translated workflows into automated pipelines through close collaboration and continuous deployment.
- Delivered ad-hoc analyses and data engineering for investor presentations and internal reporting.
- Developed NLP-powered keyword clustering combined with quantitative data, reducing work from days to minutes.

Research Scientist — National Physical Laboratory (NPL) *Sep. 2015 to Feb. 2021*

Produced a novel scaling indicator for tipping point detection in multidimensional time series data.

- Presented results for stakeholder meetings at NPL, in published articles, and at international conferences.
- Developed methods to detect tipping points in dynamical systems, working with large geospatial time series datasets and building stochastic models to test hypotheses.
- Code written in *MATLAB* and *Python*. Visualisations produced using *Matplotlib* and *XMGrace*.
- Accessing, cleaning, interpolating and correctly interpreting large, raw meteorological datasets.

Associate Lecturer — Sheffield Hallam University *Sep. 2017 to Jul. 2019*

Taught mathematics, statistics and computing courses from Foundation Degree to Masters.

- Taught *Microsoft Excel for Business* and *Intro to Computer Science*, besides maths and stats courses.
- Responsibilities included lesson preparation, course planning, marking, leading tutorials.
- Manned the "Maths Help" desk three times a week, providing support to students of all departments.

Informatics developer (internship) — UK Met Office Informatics Lab *Jun. to Aug. 2017*

Project assessing the viability of using amateur meteorological data to augment official observations.

- Accessed internal and external data via APIs using Java.
- Data cleaning and analyses in Python.
- Worked with the Informatics team and participated in daily stand-ups.

Academic background

Ph.D. Mathematics — University of Reading
Thesis: *Tipping Points and Early Warning Signals with Applications to Geophysical Data*.

- Thesis develops spectral analysis methods to detect correlations in multidimensional time series, which are used as Early Warning Signals for tipping events in stochastic dynamical systems.
- Methods applied to real geospatial time series data: hurricane paths and greening/desertification.
- Publication of three papers in respected journals:
  - *A novel scaling indicator of early warning signals* (EPL, 2018)
  - *Generalized early warning signals in multivariate and gridded data* (Chaos, 2019)
  - *Power spectrum scaling as a measure of critical slowing down* (ERL, 2022)
- Implemented mathematically derived methods numerically by discretising and linearising.
- Participation in several international conferences and workshops.
- Three-month data-focussed internship at the UK Met Office (see *Professional Exp.* section).
- EPSRC funded research in industrial partnership with NPL. Attracted additional funding from NERC.

M.Res. Mathematics — Imperial College London (*Distinction*)
Dissertation: *Mesh Generation Using Solutions of an Optimal Transport Problem*.

- Dissertation uses solutions to a linearisation of the Monge-Ampère equation to generate an adaptive, optimally distributed mesh for numerical models.
- Working in C++; monitoring CPU usage and convergence time over various scenarios.
- Participation in weekly soft skills workshops and monthly internal showcases.
- Attendance at weekly Mathematics of Planet Earth seminars hosted by the Centre for Doctoral Training.
- Completed taught courses in Python, R, Dynamical Systems, Stochastic differential equations, Bayesian Analysis and Probability.

MA Mathematics — University of Edinburgh (*First Class Hons.*)
Dissertation: *Growth and its Applications in Graph Theory*.

- Dissertation explores patterns in paths through connected digraphs.
- Wrote a program with a simple UI to investigate the properties of these digraphs: this then became a teaching aid, earning a letter of thanks from the principal.
- Requisite taught courses for the Pure Mathematics degree besides optional modules in computing and mathematics education.
- Maths outreach in local schools, museum events and university seminars.